ENTERPRISE DATA ASSESSMENT

Data Workbench

Enterprise data diagnostics, issue management, root-cause analysis and remediation

Product framework, operating journey and diagnostic reference

REPOSITORY BASELINE 0.5.2

07 SEPTEMBER 2026

ASSESS

Govern scope and calculate evidence

EXPLAIN

Review findings and investigate causes

ACT

Approve conclusions and track follow-up

Turn data-quality signals into reviewable decisions

Data quality (DQ) assessment for analytics and modeling, supported by retained evidence.

What it is

A workspace to version data, configure diagnostics and investigate findings.

How it works

Governed scope and deterministic analysis, with explicit user decisions at key handoffs.

What teams gain

Traceable evidence, consistent review and a recorded path from finding to conclusion.

5 launch-enabled diagnostics • 1 implemented but paused • 3 registered planned workflows

Expose analytical risks before they become modeling problems

The framework assesses whether data is fit for its intended analytical use.

Data can look complete

A valid file can still contain biased populations, immature outcomes or leaked predictors.

Meaning can be unclear

Sentinels, censoring and conditional fields need business context before a quality decision.

Evidence can fragment

Repeated manual checks and disconnected issue notes make an investigation hard to reproduce.

The assessment joins data context, analytical evidence and accountable review.

Build confidence through repeatable assessment and clear ownership

Intended outcomes are supported by concrete workflow capabilities.

Repeat an assessment

CONSISTENCY

Use active snapshots, frozen scope, methodology versions and reusable artifacts.

Explain a result

TRACEABILITY

Retain metrics, thresholds, exceptions, findings and user rationale together.

Coordinate follow-up

ACCOUNTABILITY

Approve an RCA conclusion; optionally assign tracked remediation to an owner.

Prioritize fitness for purpose; calibrate enterprise benefit with real operating evidence.

Use a governed framework across the assessment journey

Scope, rules and review remain explicit as data moves through the product.

01	→	02	→	03	→	04	→	05
Ingest & profile		**Configure scope**		**Run diagnostics**		**Review & investigate**		**Follow up & reassess**
Review source and context		Confirm roles and rules		Calculate retained evidence		Decide findings and RCA		Track work; launch a fresh run

TEST LAB	ISSUE & RCA	SHARED SERVICES
Manifests • engines • findings • reports	Investigation • evidence • human conclusion	Assets • knowledge • access • artifacts

Framework & knowledge: dimensions, rules, thresholds

Deterministic engines: reproducible analytical evidence

AI can advise and explain. Humans confirm meaning, findings and conclusions.

Keep framework coverage distinct from executable coverage

Six themes and eleven assessment areas organize the long-term assessment scope.

Sample & representation

Portfolio mix; historical depth; downturn and regime coverage.

Target & outcome

Target definition and label consistency.

Feature quality

Missingness; extremes; temporal leakage; behavioral stability.

Dataset monitoring

Dataset drift and degradation over time.

Pipeline integrity

Schema consistency and continuity of delivery.

Third-party controls

External-data robustness and change control.

Framework coverage includes planned and unimplemented areas; report that scope explicitly.

Give each participant a clear purpose and handoff

Business personas describe responsibilities; authorization roles are shown on the next slide.

Sponsor / product owner

Proposed: set priorities, accept rollout scope and review evidence packs.

Analytics / model team

Select target and scope; interpret separation, drift and context.

Data team

Upload assets; confirm mappings, periods and schema changes.

Diagnostic owner / SME

Proposed ownership: maintain methodology and resolve ambiguous findings.

Governance team

Edit or review knowledge; approve governed publication.

IT & remediation owners

Administer users and storage; own assigned follow-up work.

Subject-matter expert (SME): a business or analytical reviewer of meaning and evidence.

Apply the roles that the product actually enforces

Role-based access control (RBAC) governs administration and Knowledge Base (KB) permissions.

Role	Implemented authority	Boundary
user	Normal authenticated product access	No separate diagnostic-owner or remediation role
admin	Manage users and administrative actions	Does not inherit KB publication rights
kb_editor	Upload and prepare knowledge drafts	Publication requires reviewer authority
kb_reviewer	Publish or archive governed knowledge	Model output cannot publish directly

To be validated: enterprise identity, full route coverage and finer-grained asset access.

Follow the evidence from source data to an approved conclusion

The primary product journey ends with an investigation outcome; follow-up remains visible.

01
Source
Upload, profile, confirm

→

02
Assess
Scope, freeze, execute

→

03
Review
Findings and dispositions

→

04
Investigate
Evidence, hypotheses, challenge

→

05
Conclude
Approve cause or unresolved

Optional tracked remediation
Owner • priority • target date • tracked status

Manual reassessment
Correct data → upload/review → run diagnostics

RCA closure records an accepted investigation outcome; it does not certify remediation.

Prepare an active snapshot and make the run scope explicit

Each gate produces an artifact that the next stage can reference.

Stage / persona	Input → system action	Output / control
1 Ingest **Data team**	CSV, Excel or SQLite → stage and profile	Profile evidence; user reviews source
2 Prepare **Data team**	Mappings, target, periods → validate and commit	Ready active snapshot; commit gate
3 Configure **Analyst / SME**	Snapshot + diagnostic → resolve scope	Working manifest; confirm roles and thresholds
4 Execute **Analyst**	Confirmed manifest → freeze and run	Immutable manifest; progress and run result

Only ready, active snapshots are offered for Test Lab execution.

Turn calculation outputs into owned investigations

A finding becomes an issue through the diagnostic's specific disposition rule.

Stage / persona	Input → system action	Output / control
5 Analyze **Analyst / SME**	Run results → calculate and retain evidence	Metrics, charts and reports; scope disclosed
6 Create issue **Finding reviewer**	Finding + rationale → confirm or dismiss	Linked issue when confirmed; human gate
7 Assign / prioritize **Issue or action owner**	Reviewed issue → optional tracked work	Owner, priority, target date; recorded by user
8 Investigate **Analyst / SME**	Issue evidence → looks, hypotheses, challenge	Evidence-backed proposed conclusion; user review

Exception: paused D04 code auto-opens deterministic VIOLATION findings when executed.

Track remediation without overstating the closure control

Current capabilities support follow-up; corrected-data validation remains a separate activity.

Stage / persona	Available action	Artifact / control
9 Remediate **Assigned owner**	Create tracked work; apply source fix externally	Tracked record implemented; source change manual
10 Validate **Data + analytics teams**	Review a new upload and launch a new diagnostic	New snapshot/run evidence; manual reassessment
11 Close / report **RCA reviewer**	Approve identified or unresolved conclusion	RCA and managed issue close with rationale

Proposed enterprise gate: require fresh retest evidence before declaring remediation complete.

Keep run state, finding review and issue state separate

A completed execution is different from an accepted finding or a closed investigation.

RUN

draft → **running** → **done / failed**

A draft may be discarded. Frozen history is retained.

FINDING

open → **confirmed / dismissed** → **linked issue**

A review decision remains separate from the calculated result.

ISSUE

Open → **In RCA / Escalated** → **Closed**

Closure records rationale or an approved RCA conclusion.

No result is silently rewritten by a later disposition, fix handoff or new upload.

Distinguish evidence, findings and issues in every handoff

Use the same terms across diagnostics, analytics and remediation.

Evidence

WHAT WAS OBSERVED

Measured outputs and provenance: snapshot, manifest, thresholds, charts and artifacts.

Finding

WHAT NEEDS REVIEW

A rule outcome or contextual review prompt, linked to the evidence that produced it.

Issue

WHAT IS BEING INVESTIGATED

A managed record after the required creation gate, with a route into root-cause analysis.

Statistical flags are review prompts. A flagged association is not proof of a data defect.

Approve a conclusion with its supporting and limiting evidence

The visible RCA journey is Intake → Investigate → Conclusion approval.

01
Intake
Review source finding and scope

→

02
Investigate
Run looks; assess hypotheses

→

03
Challenge
Consider alternatives and limits

→

04
Approve conclusion
Identify cause or record unresolved

Approval evidence
Cause or unresolved rationale; evidence references; alternatives; limits; related failures.

Optional handoffs
Create tracked remediation. Propose reusable knowledge only when a cause is identified.

A reviewer may return the conclusion to investigation with a recorded reason.

Govern knowledge before it becomes reusable authority

Knowledge Base (KB) lifecycle and diagnostic reasoning have distinct decision gates.

01	02	03	04
Draft	**Review**	**Publish**	**Use & maintain**
Upload or propose knowledge	Inspect rules, scope and evidence	Authorized reviewer acts	Version, challenge or archive

Human control

Editors prepare; reviewers publish. AI advice requires the applicable human gate.

Versioned application

Runs retain the rules, terminology and decisions used at the time of execution.

D11 proposal publication and incorporation into a future packaged KB release are separate.

Retain the chain from snapshot to decision

The Analysis Artifact Repository (AAR) stores governed analytical evidence.

01

Asset / snapshot

Source version and profile

→

02

Frozen manifest

Scope, rules, parameters

→

03

Analytical artifacts

Metrics, payloads and lineage

→

04

Finding / issue

Rationale and linked evidence

→

05

RCA record

Evidence and approved conclusion

Captured across the chain

Actor • timestamp • snapshot ID • rule/KB version • methodology • hashes • source references

Reporting must disclose what ran, what did not run and where evidence is limited.

Separate the user workflow from the execution and evidence services

Implementation architecture supports the operating model without implying deployment certification.

USER WORKSPACE	Data Sourcing • Test Lab • Issue Management • Knowledge Base • Admin
APPLICATION SERVICES	Asset/version control • Diagnostics • RCA • Authentication • Reporting
EXECUTION & PERSISTENCE	Deterministic Python engines • SQLite state • Upload storage • AAR artifacts

AI services are configured separately; deterministic calculations retain the verdict boundary.

See which diagnostic implementations can be launched

Implementation status and register availability are intentionally separate.

ID / diagnostic	Purpose	Input → main output	Status / source
T1-D02 Target separation	Review feature–target separation	Features + target → AUC / IV evidence	Implemented • enabled D02R
T2-D04 Cross-field rule	Check governed business rules	Rows + KB rules → verdicts	Implemented • paused D04R
T2-D06 Row completeness	Find panel continuity gaps	Facility × period → gap evidence	Implemented • enabled D06R
T2-D08 Value semantics	Explain exceptional values	Fields + roles → tags and coverage	Implemented • enabled D08R
T2-D11 Directional consistency	Compare expected / observed direction	Numeric features + reference → evidence	Implemented • enabled D11R
T4-D14 Population Stability Index (PSI)	Compare two populations	Baseline + Current → PSI report	Implemented • enabled D14R

Primary users: analytics/model teams and SME reviewers; data teams prepare scope.
AUC: area under the receiver operating characteristic curve. IV: Information Value.

Keep planned workflows and supporting analytics visible

All nine registered entries are represented; supporting investigations sit beside the register.

Entry	Purpose / primary user	Input → output	Status / source
T3-D12 Label consistency	Reconcile labels / data + SME	Labels + process rules → violations	Planned; pending S08, S02 row 13
T5-D17 Maturity by vintage	Review resolved outcomes / model team	Vintage + outcome state → resolved share	Planned; pending S08, S02 row 18
T6-D20 AI-proposal workflow	Grow governed relationships / SME	Profiles + KB → reviewed proposals	Planned; pending S08, S02 row 21
SUP-MISS Missingness Mechanism	Explain missingness / analytics	Table + masks → descriptive report	Implemented; supporting S23, S24

Legend: Implemented = code exists; Planned = documented; Referenced = specification incomplete.

Retain deferred diagnostics without implying availability

These eleven workbook diagnostics are planned and absent from the current register.

ID / diagnostic	One-line purpose	User / input	Output / source
T1-D01 Post-outcome field / look-ahead present	Find information available after outcome.	Analytics + SME Features + availability dates	Availability flags S02 row 2
T1-D03 Categorical leakage (association strength / purity)	Find categories that almost reveal target.	Analytics + SME Categories + target	Purity / association S02 row 4
T2-D05 Derivation identity	Check a derived field against its formula.	Analytics + SME Recorded / derived fields	Formula discrepancy S02 row 6
T2-D07 Disguised-missing / staleness	Detect frozen or sentinel values.	Analytics + SME Panel fields + sentinels	Frozen / sentinel flags S02 row 8
T2-D09 Statistical outlier (robust-Z)	Flag improbable numeric extremes.	Analytics + SME Material numeric fields	Robust-Z candidates S02 row 10
T2-D10 Boundary / pile-up detection	Measure mass at declared bounds.	Analytics + SME Bounded numeric field	Boundary share S02 row 11

Planned • not registered • no current launch, automatic issue creation or diagnostic report.

Retain deferred diagnostics without implying availability

These eleven workbook diagnostics are planned and absent from the current register.

ID / diagnostic	One-line purpose	User / input	Output / source
T3-D13 Target-rate break (change-point)	Find structural target-rate shifts.	Analytics + SME Target + dated definitions	Change-point evidence S02 row 14
T4-D15 KS distance	Measure distance from reference.	Analytics + SME Sample + reference	KS distance S02 row 16
T4-D16 Segment coverage	Find absent or thin segments.	Analytics + SME Sample + reference segments	Absent / thin segments S02 row 17
T5-D18 Resolved-only vs all differential	Compare resolved-only vs all estimates.	Analytics + SME Resolved / full outcomes	Estimate differential S02 row 19
T5-D19 Seasoning / maturity profile	Assess outcome seasoning.	Analytics + SME Vintage + age + outcome	Maturity curve S02 row 20

Planned • not registered • no current launch, automatic issue creation or diagnostic report.

Make broader framework references explicit

These references extend the framework; they are not additional executable register rows.

Reference / purpose	Input	Intended output	Status
R01 Formal MCAR / MNAR testing	Numeric missingness, target and semantics	Formal mechanism assessment	Planned / deferred; descriptive supporting analysis is available
R02 Downturn / regime identification	Macro/outcome history and stress context	Regime and stress-coverage evidence	Referenced; no sufficient implementation contract
R03 Feature drift over vintages	Dated snapshots and training baseline	Feature drift over time	Partially implemented: two-snapshot PSI; monitoring workflow planned
R04 Delivery drift / degradation	Prior loads, schema, volume and null profiles	Load-to-load degradation signals	Partially implemented: version/profile comparison; full monitoring planned

MCAR: missing completely at random. MNAR: missing not at random. Both require cautious interpretation.

Make broader framework references explicit

These references extend the framework; they are not additional executable register rows.

Reference / purpose	Input	Intended output	Status
R05 Survival / Kaplan–Meier analysis	Observed durations and event/censor flags	Survival and window analysis	Referenced; not registered or implemented as a diagnostic
R06 Portfolio migration / acceptance funnel	Population states and selection history	Selection-bias evidence	Referenced; insufficient method and workflow detail
R07 Tail / cluster investigation	Material numeric fields and context	Extreme-value behavior evidence	Referenced; legacy tail test retired
R08 Snapshot / timestamp lineage checks	As-of dates, events and transformations	Temporal integrity evidence	Referenced; overlaps deferred D01, no separate executor
R09 Data latency / feed monitoring	Arrival times, feed expectations and changes	Pipeline timing and control evidence	Referenced; no current monitored-alert workflow

MCAR: missing completely at random. MNAR: missing not at random. Both require cautious interpretation.

Account for the retired fourteen-test library

The historical README describes an older product; its registry is empty in current code.

Historical diagnostic	Current location / successor	Current disposition
PSI	Replaced by current D14	Implemented successor; historical registry retired
Leakage detection (single-feature AUC)	Replaced by D02	Implemented successor; historical registry retired
Plausibility rule check	Absorbed into D04 KB rules	Implemented successor paused
Date-ordering / event-sequence	Absorbed into D04 KB rules	Implemented successor paused
Completeness / missing-rate profile	Profiling input; different D06 panel method	Implemented profiling; legacy diagnostic retired
Post-outcome field inventory	Deferred D01	Planned successor; retired legacy test
Maturity profile analysis	Deferred D19	Planned successor; retired legacy test

Retired catalog entries have no current launch. Current successors are covered in dedicated sections.

Account for the retired fourteen-test library

The historical README describes an older product; its registry is empty in current code.

Historical diagnostic	Current location / successor	Current disposition
MCAR test (Little’s)	Formal mechanism test deferred	Planned; supporting missingness is distinct
Missingness-vs-target (MNAR)	Formal mechanism inference deferred	Planned; association is not proof of MNAR
Robust z-score outlier (MAD)	Deferred D09	Planned successor; retired legacy test
Key uniqueness check	Profiling precondition	Implemented precondition; no standalone diagnostic
Tail analysis	Legacy test deleted	Referenced history; no current executor
Correlation stability analysis	Legacy test deleted	Referenced history; D11 is a different method
Drift decomposition	Legacy test deleted	Referenced history; full Plan B workflow absent

Retired catalog entries have no current launch. Current successors are covered in dedicated sections.

Implemented • enabled

Spot features whose target separation deserves review

Single-feature target separation

Problem

A feature can look implausibly predictive, or separate the target poorly on its own.

Primary users

Analytics and model-development teams, supported by an SME reviewer.

When to use it

A ready model dataset with a confirmed target and eligible independent variables.

Feature-level separation categories and candidate findings with retained statistical and binning evidence.

Implemented • enabled

Follow the governed processing path

Single-feature target separation

INPUTS One ready table; governed target type/event class; selected features; binning constraints and thresholds.

01
Resolve target
Confirm target route and feature eligibility

→

02
Calculate ROC
Receiver operating characteristic evidence

→

03
Build bins
Generate fine / coarse bins and IV

→

04
Classify
Apply category precedence and candidate rules

→

05
Retain evidence
Persist feature artifacts and report

Uses shared binning and AAR. Confirmed target is required; no AI adjudication component.

Implemented • enabled

Read the metrics alongside scope and limitations

Single-feature target separation

ROC / AUC / Gini

Receiver operating characteristic (ROC), area under the curve (AUC) and Gini quantify univariate separation.

WOE / IV and bins

Weight of evidence (WOE), Information Value (IV), fine/coarse definitions, counts and optimizer warnings.

Governed report

Feature metrics, category rationale, candidate reasons, dispositions and linked source artifacts.

Category precedence: Suspicious → Strong → Medium → Weak. Categories do not prove leakage.

Implemented • enabled

Confirm the scope, launch once and review final evidence

Single-feature target separation

01
Choose snapshot
Analyst selects the ready active asset

→

02
Confirm scope
Review target, features and constraints

→

03
Freeze & run
Persist scope before execution

→

04
Monitor
Read progressive feature previews

→

05
Review final
Open findings, artifacts and report

Success / completion

Complete or partial feature evidence remains visible. Final findings become available after run finalization.

Failure / blocked scope

Invalid/action-required inputs and per-feature ROC/binning errors are explicit; previews are not final results.

Implemented • enabled

Carry the finding into a controlled investigation

Single-feature target separation

Step	Implemented behavior / proposed follow-up
Finding	Potential leakage or poor discrimination becomes a candidate finding.
Issue gate	Human confirms or dismisses with rationale; confirmed finding links to Issue Management.
RCA	Review feature timing, target construction and retained ROC/IV/bin evidence. Source-evidence tools support bin review.
Remediation / retest	Suggested: correct construction or review feature use; manually upload corrected data and reassess.
Closure	Reviewer approves root cause identified or unresolved with evidence; optional follow-up stays separate.

Manual / proposed: source correction and fresh-data retest. RCA approval does not prove a fix.

Implemented • paused

Check whether rows satisfy governed business rules

Cross-field business rule

Problem

Rows may violate cross-field relationships, domain bounds or event-ordering rules.

Primary users

Data-quality analysts and domain SMEs; KB editors and reviewers maintain governed rules.

When to use it

Code supports a ready asset with published, bound rules in the selected use case. New launches are paused.

PASS, VIOLATION or NOT-APPLICABLE per rule, with exceptions, rates and bounded evidence.

Implemented • paused

Follow the governed processing path

Cross-field business rule

INPUTS Published and bound KB rules; selected use case; resolved role-to-column bindings; source tables and tolerances.

01
Select rules
Published, bound and in scope

→

02
Resolve roles
Map required business fields

→

03
Apply scope
Conditions, exceptions and missingness

→

04
Evaluate rules
Count violations over evaluated rows

→

05
Persist & hand off
Store verdicts; synchronize violations

Availability gate: workflow_pending. Engine and report code exist; current configuration blocks new launches.

Implemented • paused

Read the metrics alongside scope and limitations

Cross-field business rule

Rule-level accounting

Evaluated, excepted and skipped row counts; violation count/rate; resolved roles and rule text.

Pattern evidence

Bounded examples and segment concentration support localizing a rule violation.

Structured report

Dataset rollup, severity, parameter sources and NOT-APPLICABLE reasons; PDF/text rendering code.

Decision rule: violation rate ≤ tolerance → PASS; otherwise VIOLATION.

Implemented • paused

Confirm the scope, launch once and review final evidence

Cross-field business rule

01

Inspect coverage

Current card shows workflow pending

→

02

Resolve scope

When re-enabled: use case and tables

→

03

Review rules

Confirm roles, exclusions and tolerances

→

04

Freeze & execute

Supported code path; unavailable now

→

05

Read report

Historical evidence remains reviewable

Success / completion

Implemented path produces complete rule accounting and hands VIOLATION findings to issue synchronization.

Failure / blocked scope

Current launches are refused. In the code path, missing scope yields reasons; execution errors are sanitized.

Implemented • paused

Carry the finding into a controlled investigation

Cross-field business rule

Step	Implemented behavior / proposed follow-up
Finding	A deterministic VIOLATION is the finding condition after exceptions and tolerance.
Issue gate	Implemented runner auto-opens issues for violations when executed; PASS/NOT-APPLICABLE do not.
RCA	Inspect the governed rule, bindings, exceptions and segment evidence; verify whether data or rule design is wrong.
Remediation / retest	Suggested: correct source logic or submit a governed rule change; retest manually after approval and re-enablement.
Closure	Reviewer approves root cause identified or unresolved with evidence; optional follow-up stays separate.

Manual / proposed: source correction and fresh-data retest. RCA approval does not prove a fix.

Implemented • enabled

Find missing facility-period rows inside observed lifespans

Row-completeness reconciliation

Problem

Panel gaps and duplicate keys can distort longitudinal analysis even when received rows are populated.

Primary users

Data teams and analytics reviewers who understand facility identifiers, period grain and optional segments.

When to use it

One ready panel table with a facility ID, reporting period, confirmed grain and continuity floor.

Six deterministic rule results and continuity evidence, with human review before an issue is created.

Implemented • enabled

Follow the governed processing path

Row-completeness reconciliation

INPUTS One panel table; facility_id and period bindings; grain; optional segment; continuity floor (default 95%).

01
Validate keys
Check identifiers and parse periods

→

02
Define spans
First-to-last observed period per facility

→

03
Build grid
Required facility-period pairs inside spans

→

04
Reconcile
Detect gaps, duplicates and segment gaps

→

05
Summarize
Retain six rule results and coverage

Observed-span method only: cannot detect wholly absent facilities or missing rows beyond observed endpoints.

Implemented • enabled

Read the metrics alongside scope and limitations

Row-completeness reconciliation

Continuity metric

Received required facility-period pairs divided by required pairs within observed spans.

Six rule outcomes

Identifiers; period sequence; rows by period; duplicate pairs; facility gaps; segment gaps.

Decision evidence

Masked failed-row examples, rule rationales, affected populations and PDF/text report.

A high portfolio coverage rate can coexist with a failed period or segment rule.

Implemented • enabled

Confirm the scope, launch once and review final evidence

Row-completeness reconciliation

01	02	03	04	05
Select table	**Confirm roles**	**Set method**	**Freeze & run**	**Review rules**
Use one ready panel snapshot	Facility ID, period and optional segment	Review grain and continuity floor	Use the frozen observed-span scope	Inspect findings and governed report

Success / completion

Six rule outcomes, masked evidence and continuity metrics are retained. Optional segment can be NOT-APPLICABLE.

Failure / blocked scope

Input/period incompatibilities are reported; failed rules await human review and do not auto-create issues.

Implemented • enabled

Carry the finding into a controlled investigation

Row-completeness reconciliation

Step	Implemented behavior / proposed follow-up
Finding	A VIOLATION in any of the six rules becomes a reviewable finding.
Issue gate	Human confirms or dismisses failed-rule findings with rationale before issue creation.
RCA	Review invalid keys, calendar gaps, duplicate type and affected period/segment concentrations.
Remediation / retest	Suggested: reconcile source deliveries or deduplicate; upload corrected panel and rerun for fresh evidence.
Closure	Reviewer approves root cause identified or unresolved with evidence; optional follow-up stays separate.

Manual / proposed: source correction and fresh-data retest. RCA approval does not prove a fix.

Implemented • enabled

Explain exceptional values before treating them as defects

Value-semantics classification

Problem

Blanks, sentinels, static values and conditional fields can have legitimate business meaning.

Primary users

Data and model teams, with an SME confirming semantic roles when exact evidence is insufficient.

When to use it

A ready table with selected fields, confirmed role coverage and any required runtime declarations.

Non-destructive cell tags, coverage gaps and grouped candidate findings; source values stay unchanged.

Implemented • enabled

Follow the governed processing path

Value-semantics classification

INPUTS Selected fields; confirmed canonical roles; versioned KB/terminology; declared sentinels, dates or panel rules.

01
Bind roles
Exact-first matching; confirm other proposals

→

02
Route rules
Expand roles and check prerequisites

→

03
Classify cells
Run deterministic KB primitives

→

04
Resolve precedence
Retain one winning tag per cell

→

05
Publish evidence
Tags, assessment ledger and report

General context is valid. Credit-risk use context is optional and does not enable or suppress rules.

Implemented • enabled

Read the metrics alongside scope and limitations

Value-semantics classification

Three governed tags

CENSORED • STALE_FROZEN • NOT_APPLICABLE. Tags explain a cell; they do not edit it.

Coverage outcomes

NO_TAG • UNCLASSIFIED • UNSCOPED. No tag does not certify validity.

Four AAR artifacts

Bindings JSON; sparse tags Parquet; assessment ledger Parquet; structured report JSON.

Optional AI reviews bounded metadata; users confirm non-exact roles; deterministic rules create tags.

Implemented • enabled

Confirm the scope, launch once and review final evidence

Value-semantics classification

01
Acknowledge About
Understand tags and limitations

→

02
Data & context
Select table; context is optional

→

03
Confirm bindings
Review scope and semantic roles

→

04
Rules & launch
Review declarations; freeze scope

→

05
Review results
Tags, coverage, report and dispositions

Success / completion
Deterministic tags and exhaustive routed coverage publish to AAR, with authenticated reports and inference disclosure.

Failure / blocked scope
Unacknowledged intro or no usable confirmed role coverage blocks freeze. Missing optional inputs yield UNSCOPED.

Implemented • enabled

Carry the finding into a controlled investigation

Value-semantics classification

Step	Implemented behavior / proposed follow-up
Finding	Stale/frozen patterns and populated not-applicable values can form grouped issue candidates.
Issue gate	User confirms or dismisses candidates with rationale. No issue or RCA is started automatically.
RCA	Use role bindings, rule reasons, precedence and coverage evidence; review semantics before alleging a defect.
Remediation / retest	Suggested: correct declarations or source capture when justified; run a new assessment after change.
Closure	Reviewer approves root cause identified or unresolved with evidence; optional follow-up stays separate.

Manual / proposed: source correction and fresh-data retest. RCA approval does not prove a fix.

Implemented • enabled

Compare observed relationships with governed expectations

Directional / monotonic consistency

Problem

A numeric feature can move against the expected risk direction, or show weak or conflicting evidence.

Primary users

Analytics and domain SMEs reviewing economically meaningful numerical relationships.

When to use it

Ready data with a saved target; confirm reference type, risk orientation and selected numeric features.

Expected-versus-observed conclusions, component evidence and contextual review findings.

Implemented • enabled

Follow the governed processing path

Directional / monotonic consistency

INPUTS Numeric features; target or approved run-level substitute; risk orientation; expected directions; evidence floors.

01
Set expectation
Exact KB or explicit human confirmation

→

02
Measure components
Binned shape, Spearman and regression

→

03
Classify direction
Consensus plus applicability/data guards

→

04
Compare
Orient expected risk direction to reference

→

05
Retain & review
Feature artifacts, charts and report

First run is overall-only. A later run can add one accepted segment using the shared PSI split builder.

Implemented • enabled

Read the metrics alongside scope and limitations

Directional / monotonic consistency

Three empirical inputs

Broad binned averages, Spearman rank correlation and standardized univariate regression.

Review conclusions

Agreement, review recommended, weak/no relationship, conflicting evidence or insufficient data.

Governed evidence

Feature-level directionality_evidence and run-level directionality_report; charts and JSON evidence.

Pearson correlation is displayed for context; it does not vote in the direction class.

Implemented • enabled

Confirm the scope, launch once and review final evidence

Directional / monotonic consistency

01
Resume or create
Restore a compatible working draft

→

02
Confirm reference
Set type, event class and orientation

→

03
Select features
Resolve each expected direction

→

04
Freeze & run
Atomic manifest and optional proposal

→

05
Review evidence
Overall first; optional later segment

Success / completion
Retained confirmations can be reused on compatible reruns; new overall/segment evidence is computed.

Failure / blocked scope
Unresolved scope blocks freeze; atomic freeze failure leaves the draft. Feature failures remain explicit.

Implemented • enabled

Carry the finding into a controlled investigation

Directional / monotonic consistency

Step	Implemented behavior / proposed follow-up
Finding	REVIEW_RECOMMENDED, CONFLICTING_EVIDENCE or WEAK_OR_NO_RELATIONSHIP creates contextual review findings.
Issue gate	An SME promotes or dismisses with rationale; observed evidence remains immutable.
RCA	Compare expected direction, target orientation, cleaning counts, component evidence and segment context.
Remediation / retest	Suggested: investigate mapping, population or business change; propose reviewed KB updates where warranted.
Closure	Reviewer approves root cause identified or unresolved with evidence; optional follow-up stays separate.

Manual / proposed: source correction and fresh-data retest. RCA approval does not prove a fix.

Implemented • enabled

Locate distribution changes between governed populations

Population Stability Index (PSI)

Problem

A sample or later snapshot can differ materially from its intended baseline population.

Primary users

Analytics, portfolio and model teams reviewing representativeness or population shift.

When to use it

One snapshot split into Baseline/Current, or two compatible snapshots; a target is optional.

PSI per feature, reconciled bin counts, contributions and contextual findings for SME review.

Implemented • enabled

Follow the governed processing path

Population Stability Index (PSI)

INPUTS Baseline and Current populations; compatible features; optional governed target; approved frozen bin definitions.

01	02	03	04	05
Define populations	**Route binning**	**Review & freeze**	**Calculate PSI**	**Interpret**
One split or two snapshots	Target-free or D02 target-aware bins	Approve rules applied to both sides	Compare proportions in identical bins	Stable / watch / investigate with evidence

Target-aware binning uses D02. Two-snapshot reuse requires an exact, explicitly promoted universal definition.

Implemented • enabled

Read the metrics alongside scope and limitations

Population Stability Index (PSI)

Population accounting

Baseline / Current counts and shares, split predicate or snapshot lineage, missing and special bins.

PSI contribution table

Each bin's contribution and reconciled totals; Current-only categories are retained as UNSEEN.

Reviewable outputs

Feature evidence, frozen bin lineage, result classifications and run reports.

Governed defaults: stable < 0.10; watch ≥ 0.10; investigate ≥ 0.25.

Implemented • enabled

Confirm the scope, launch once and review final evidence

Population Stability Index (PSI)

01
Choose populations
One snapshot split or two snapshots

→

02
Select route
Use governed target or target-free

→

03
Prepare bins
Read Baseline; generate or exact-reuse

→

04
Approve & execute
Freeze common bins and run PSI

→

05
Review final
Contributions, findings and report

Success / completion
Server-owned work continues across browser reconnects. Buffered previews remain preliminary until finalization.

Failure / blocked scope
Incompatible scope, invalid cuts or non-reconciling bins are rejected; unavailable feature evidence is explicit.

Implemented • enabled

Carry the finding into a controlled investigation

Population Stability Index (PSI)

Step	Implemented behavior / proposed follow-up
Finding	Watch/investigate population shifts create contextual findings; they are not automatic data violations.
Issue gate	SME confirms or dismisses with rationale before issue creation and the existing RCA handoff.
RCA	Review which bins drive PSI, population selection, missing/special values and any legitimate business shift.
Remediation / retest	Suggested: adjust a justified sample/baseline or correct data; approve fresh bins where required and rerun.
Closure	Reviewer approves root cause identified or unresolved with evidence; optional follow-up stays separate.

Manual / proposed: source correction and fresh-data retest. RCA approval does not prove a fix.

Implemented • supporting

Explain missingness patterns without asserting causality

Missingness Mechanism is implemented beside the nine-diagnostic register.

Purpose

Identify descriptive structure in missingness and distinguish weak evidence from untestable scope.

Inputs

Ready active table; missing-value codes; period context; optional tolerances and analysis parameters.

Outcome

Immutable missingness report and above-tolerance observations for human disposition.

Missing at random terminology is a claim boundary: the analysis does not prove a mechanism.

Freeze the missingness scope and retain descriptive evidence

Co-missingness, structural breaks and shallow-tree explanations are preserved separately.

01
Freeze scope
Snapshot, codes, parameters and seed

→

02
Build masks
Missingness and co-missing blocks

→

03
Apply gates
Seven classification gates and guards

→

04
Explain patterns
Shallow trees where analytically valid

→

05
Store report
Immutable missingness_report artifact

Read the report
Column rates, descriptive patterns, structural period breaks and untestable cases.

Interpret cautiously
No pattern found is distinct from an untestable analysis; association does not establish cause.

Review observations before starting an issue or RCA

Use Test Lab's supporting-investigation entry point on a ready snapshot.

Stage	Control and output
Launch	Select table and parameters; backend validates ready active scope and freezes the manifest.
Complete / fail	Persist an immutable report; surface readiness and execution failures explicitly.
Finding → issue	User confirms or dismisses an above-tolerance observation with a recorded reason.
RCA and follow-up	Review capture patterns and source context; manually correct/reassess where justified.
Closure	Approve a supported or unresolved RCA conclusion; retain the original report as evidence.

This analysis complements value-semantics review; automatic D08 tag consumption needs validation.

Planned • workflow pending

Check label consistency against the business process

Label-consistency rule

Purpose & user

Reconcile internally inconsistent label fields using governed process rules. Data and domain SMEs would review conflicts before accepting an issue.

Inputs & approach

Target/label fields, event/process evidence and the applicable KB rules. Planned method: deterministic violation count after documented rule exceptions.

Launch & downstream

No current launch. Proposed: review evidence, decide a finding, then use issue/RCA and manual follow-up.

Planned capability: thresholds, workflow controls, report schema and acceptance remain to be validated.

Planned • workflow pending

Review how fully each vintage’s outcomes have resolved

Resolution / maturity rate by vintage

Purpose & user

Identify vintages whose unresolved outcomes may bias analytical conclusions. Model and portfolio teams would distinguish incomplete observation from a defect.

Inputs & approach

Vintage identifier, outcome state and a definition of resolved versus open. Planned method: resolved count / total count by vintage against a tunable floor.

Launch & downstream

No current launch. Proposed: review evidence, decide a finding, then use issue/RCA and manual follow-up.

Planned capability: thresholds, workflow controls, report schema and acceptance remain to be validated.

Planned • workflow pending

Expand governed relationships through an SME-reviewed loop

AI-proposal workflow

Purpose & user

Propose relationships absent from the existing KB and retain human acceptance. SMEs would review candidates; confidence ranks proposals and is not a data metric.

Inputs & approach

Profile metadata, existing KB, candidate provenance and an SME review decision. Planned sequence: propose → validate → persist → route into a Test 2 engine.

Launch & downstream

No current launch. Proposed: review evidence, decide a finding, then use issue/RCA and manual follow-up.

Planned capability: thresholds, workflow controls, report schema and acceptance remain to be validated.

Planned • deferred

Check whether feature information arrives after the outcome

Post-outcome field / look-ahead present

Purpose & inputs

Features, availability/as-of dates and outcome timing. Is a field known only after the outcome (or a look-ahead construction) in the feature set?

Approach & output

field availability vs as-of date. Availability flags requiring confirmation of actual feature timing.

Launch & issue journey

Not registered or launchable. Proposed: SME reviews evidence, confirms any issue and routes justified work into RCA.

Deferred scope: methodology, prerequisites, workflow and evidence contract require acceptance.

Planned • deferred

Review category-level target purity with a volume guard

Categorical leakage (association strength / purity)

Purpose & inputs

Categorical features, target and minimum category volumes. Does a category value predict the outcome almost deterministically (a proxy for the target)?

Approach & output

per-category outcome rate vs base rate; Cramer's V; Information Value (alt: mutual information). NOT a chi-square significance test - significance over-fires at large n; use effect size.. Volume-gated purity / effect-size evidence; significance alone is insufficient.

Launch & issue journey

Not registered or launchable. Proposed: SME reviews evidence, confirms any issue and routes justified work into RCA.

Deferred scope: methodology, prerequisites, workflow and evidence contract require acceptance.

Planned • deferred

Reconcile recorded derived values with their formulas

Derivation identity

Purpose & inputs

Recorded derived field, formula inputs and numerical tolerance. Does a derived field equal its formula?

Approach & output

abs(recorded - derived). Recorded-versus-derived discrepancy and rule evidence.

Launch & issue journey

Not registered or launchable. Proposed: SME reviews evidence, confirms any issue and routes justified work into RCA.

Deferred scope: methodology, prerequisites, workflow and evidence contract require acceptance.

Planned • deferred

Identify fields that appear frozen or disguised as missing

Disguised-missing / staleness

Purpose & inputs

Segment-period panel, declared sentinels and expected variation. Is a normally-varying field frozen/sentinel over a segment-period?

Approach & output

within-segment variance over time; sentinel share. Frozen/sentinel evidence; overlap with D08 tags does not activate D07.

Launch & issue journey

Not registered or launchable. Proposed: SME reviews evidence, confirms any issue and routes justified work into RCA.

Deferred scope: methodology, prerequisites, workflow and evidence contract require acceptance.

Planned • deferred

Screen material numeric fields for robust outlier candidates

Statistical outlier (robust-Z)

Purpose & inputs

Material numeric fields, semantics exclusions and robust scale. Improbable extreme values in a material field's tail.

Approach & output

robust z = 0.6745*(x-median)/MAD (alt: IQR fence, isolation forest). Robust-Z candidate flags; genuine tails still require contextual review.

Launch & issue journey

Not registered or launchable. Proposed: SME reviews evidence, confirms any issue and routes justified work into RCA.

Deferred scope: methodology, prerequisites, workflow and evidence contract require acceptance.

Planned • deferred

Look for implausible concentration at declared boundaries

Boundary / pile-up detection

Purpose & inputs

Bounded field, declared limits and boundary-share threshold. Implausible mass at a bounded field's min/max.

Approach & output

share of mass at boundary. Boundary concentration evidence; scope and materiality require definition.

Launch & issue journey

Not registered or launchable. Proposed: SME reviews evidence, confirms any issue and routes justified work into RCA.

Deferred scope: methodology, prerequisites, workflow and evidence contract require acceptance.

Planned • deferred

Locate target-rate breaks around definition changes

Target-rate break (change-point)

Purpose & inputs

Dated target-rate series and label-definition change dates. Does the target rate shift over time for definitional reasons?

Approach & output

change-point / structural-break statistic on target-rate series. Change-point candidates aligned to documented definition dates.

Launch & issue journey

Not registered or launchable. Proposed: SME reviews evidence, confirms any issue and routes justified work into RCA.

Deferred scope: methodology, prerequisites, workflow and evidence contract require acceptance.

Planned • deferred

Compare sample and reference distributions using KS distance

KS distance

Purpose & inputs

Numeric sample and reference distributions. Distributional difference vs reference (complements PSI).

Approach & output

Kolmogorov-Smirnov statistic. Distribution distance, interpreted alongside sample size and PSI.

Launch & issue journey

Not registered or launchable. Proposed: SME reviews evidence, confirms any issue and routes justified work into RCA.

Deferred scope: methodology, prerequisites, workflow and evidence contract require acceptance.

Planned • deferred

Expose absent or thin population segments

Segment coverage

Purpose & inputs

Reference and sample segment keys and population counts. Is any live-population segment absent/thin in the sample?

Approach & output

presence & share by segment. Absent or underrepresented segments and their shares.

Launch & issue journey

Not registered or launchable. Proposed: SME reviews evidence, confirms any issue and routes justified work into RCA.

Deferred scope: methodology, prerequisites, workflow and evidence contract require acceptance.

Planned • deferred

Quantify the impact of resolved-only outcome selection

Resolved-only vs all differential

Purpose & inputs

Resolved flag, outcomes and comparable full/resolved samples. How much the estimate changes on resolved-only vs all.

Approach & output

delta(estimate resolved-only vs all). Difference in analytical estimates attributable to resolved-only selection.

Launch & issue journey

Not registered or launchable. Proposed: SME reviews evidence, confirms any issue and routes justified work into RCA.

Deferred scope: methodology, prerequisites, workflow and evidence contract require acceptance.

Planned • deferred

Assess whether vintages have had time to mature

Seasoning / maturity profile

Purpose & inputs

Origination/vintage, months-on-book and cumulative outcomes. Are recent vintages too immature for the outcome horizon?

Approach & output

cumulative outcome by months-on-book (seasoning curve). Seasoning curves supporting a use-case-specific maturity decision.

Launch & issue journey

Not registered or launchable. Proposed: SME reviews evidence, confirms any issue and routes justified work into RCA.

Deferred scope: methodology, prerequisites, workflow and evidence contract require acceptance.

Reconcile the inventory without double-counting capabilities

Inventory counts refer to named diagnostic/workflow records, not every statistical technique.

Inventory class	Count	Current treatment
Implemented, launch-enabled	5	D02, D06, D08, D11, D14
Implemented, launch-paused	1	D04; code/UI/tests exist, register blocks launch
Registered planned workflows	3	D12, D17, D20
Numbered deferred diagnostics	11	D01, D03, D05, D07, D09, D10, D13, D15, D16, D18, D19
Supporting investigation	1	Missingness Mechanism; separate from the register
Framework / historical references	9 / 14	Reference groups and retired names; not additional live rows

6 implemented packages + 14 planned numbered entries = the 20-row workbook diagnostic inventory.

Use repository fixtures as examples, with their provenance visible

Synthetic D06 regression fixture: continuity gaps, duplicates and invalid keys.

Facility	January 2025	February 2025	March 2025
A	Present twice	Missing	Present
B	Present	Missing	Present
Invalid row	Blank facility / invalid period	Excluded from valid-key grid	Retained as invalid evidence

6	required pairs
4	received pairs
2	missing pairs
4	primary issue instances

Synthetic test fixture • no client result • 1 surplus duplicate + 1 invalid row retained separately.

Assess the supplied sources at their proper authority level

All four files were read; current source code resolves conflicting historical statements.

Source	Coverage reviewed	Use in this deck
DQ Framework workbook	5 sheets; framework, model-family mapping, plan and tests	Purpose, taxonomy, intended users and future scope
MVP test plan workbook	5 sheets; Plan A/B, diagnostics, coverage and decisions	20-row inventory and deferred-method intent
DataQuality_Framework.pptx	5 slides; text, shapes, theme/layout and available notes	Historical product framing; lower priority than code
AI and Accelerators_Risk.pptx	4 slides; slide 4 design inspected in detail	Cream cards, coral ordinals, dark bands, Funnel Sans

Source files remain unchanged. Native rendering limitations of the supplied decks are documented.

Show implemented controls beside the remaining enterprise work

The presentation is enterprise-facing; readiness for a specific enterprise deployment needs validation.

Implemented / evidenced	Planned, proposed or to be validated
Versioned sourcing, profiles, ready snapshots	Automated source-system remediation
Five enabled diagnostics; one implemented paused	D04 re-enablement; D12/D17/D20 acceptance
Human finding decisions and approved RCA conclusion	Enforced fresh-data retest before remediation completion
Admin, KB editor and KB reviewer role gates	Enterprise identity and comprehensive route/asset authorization
AAR lineage and exact compatible reuse	Full scheduled drift/degradation monitoring
D08 semantic artifacts and descriptive missingness	Uniform downstream consumption of semantic tags

Product-owner review should confirm availability, methodology defaults and the enterprise closure policy.

Resolve source conflicts in favor of the current implementation

Historical documents are retained as evidence, with their superseded statements called out.

Conflict / assumption	Treatment in the presentation
Product naming varies	Use requested “Data Workbench”; record 0.5.2 repository baseline
Executable counts differ across documents	Use current JSON register and code dispatch: five enabled, D04 paused
Older D06/D08/D11 method descriptions differ	Use production engines and package READMEs
RCA design contains older fix/rerun sequence	Use current conclusion-approval UI; separate manual remediation
Reference includes agents, health score and monitoring	Reuse visual patterns only; omit unsupported capability claims

No live deployment, real customer results or release acceptance was inferred from static source review.

Use consistent language for data, analysis and workflow

Key abbreviations and decision terms used throughout the deck.

Term	Meaning	Term	Meaning
DQ	Data quality	RCA	Root-cause analysis
SME	Subject-matter expert	RBAC	Role-based access control
KB	Knowledge Base	AAR	Analysis / Analytics Artifact Repository
ROC / AUC	Receiver operating characteristic / area under its curve	IV / WOE	Information Value / weight of evidence
PSI	Population Stability Index	PD / LGD / EAD	Probability of default / loss given default / exposure at default
MCAR / MAR / MNAR	Missing completely at random / at random / not at random	Snapshot / manifest	Versioned data state / frozen record of run scope

Trace major claims to their source families (1/3)

Every slide also carries file-level and section/function references in its speaker notes.

Source key	Claim family	Repository location / document
S05–S07	Product / workflow / technical baseline	README.md; USER_GUIDE.md; TSD.md
S08–S13	Availability and execution gates	dq_framework_data.json; diagnostic register / dispatch / readiness; run_state.py
S14–S16	Issue, RCA and remediation boundaries	ai/v2/issues.py; domains/rca/service.py; RcaCase.jsx
S17–S20	Roles and knowledge controls	routers/admin.py; auth.py; kb.py; RCA contracts
S21–S24	Evidence, ingestion and supporting analytics	AAR domain; sourcing/profiling architecture; missingness

Speaker notes contain the full source paths and the specific contract, function or worksheet reference.

Trace major claims to their source families (2/3)

Every slide also carries file-level and section/function references in its speaker notes.

Source key	Claim family	Repository location / document
D02R/M/X/E/C	D02 method, scope, evidence and workflow	t1_d02_feature_target_separation/
D04R/M/X/E	D04 engine and paused workflow	t2_d04_cross_field_business_rule/
D06R/M/X/E/T	D06 continuity method and synthetic test	t2_d06_row_completeness/
D08R/M/X/E/K	D08 roles, tags, coverage and artifacts	t2_d08_value_semantics/; KB v0.2
D11R/M/X/E/K	D11 methodology and expected risk knowledge	t2_d11_directional_monotonic_consistency/
D14R/M/X/E/T	D14 populations, bins, PSI and tests	t4_d14_population_stability/

Speaker notes contain the full source paths and the specific contract, function or worksheet reference.

Trace major claims to their source families (3/3)

Every slide also carries file-level and section/function references in its speaker notes.

Source key	Claim family	Repository location / document
S01 / S02	Framework and 20-row inventory	Supplied framework and MVP workbooks
S03 / S04	Historical framing and visual design	Supplied presentations; reference slide 4
S25 / S26 / S34	Retired library and successor mapping	dq_tests/registry.py; 01-disposition.md; historical README
S27 / S28	Personas and older framework metadata	Product specification; 00-framework.md
S29 / S30	Artifact schemas, integrity and reuse	domains/aar/types.py; repository.py
S31–S33	Sourcing, coverage reporting and tests	routers/sourcing.py; ScorePanel.jsx; missingness tests

Speaker notes contain the full source paths and the specific contract, function or worksheet reference.

Navigate from a presentation section to its evidence

The slide range and source keys form a concise traceability index.

Slides	Section / diagnostic	Primary source keys
1–4	01 / Product	S05, S06, S04, S01, S08, S03, S21 + notes
5–6	01 / Framework	S04, S06, S08, S15, S01, S23
7	01 / People	S01, S27, S06, S16
8	01 / Access	S17, S18, S19, S20
9–12	02 / Journey	S06, S16, S15, S22, S11, S13, D02X + notes
13	02 / Lifecycle	S13, D06X, S14, S15
14	02 / Review	D02R, D11R, D14R, S14

Navigate from a presentation section to its evidence

The slide range and source keys form a concise traceability index.

Slides	Section / diagnostic	Primary source keys
15	02 / Root-cause analysis	S16, S15
16	02 / Knowledge	S06, S19, D08R, D11R, S16
17	02 / Traceability	S21, S22, S30, D08R, D11R, S15
18	02 / Platform	S05, S07, S21, S22, D14R
19–22	03 / Diagnostic catalog	S08, S10, D02R, D04R, D06R, D08R, D11R + notes
23–24	03 / Extended catalog	S01, S02, S22, D14R, S25
25–26	03 / Historical catalog	S34, S25, S26

Navigate from a presentation section to its evidence

The slide range and source keys form a concise traceability index.

Slides	Section / diagnostic	Primary source keys
27–31	04 / T1-D02	D02R, D02E, D02C, D02M, D02X, S08
32–36	04 / T2-D04	S08, D04R, D04E, D04M, D04X, S14
37–41	04 / T2-D06	D06R, D06E, D06M, D06X, D06T, S08
42–46	04 / T2-D08	D08R, D08M, D08E, D08X, D08K, S08
47–51	04 / T2-D11	D11R, D11M, D11E, D11X, D11K, S08
52–56	04 / T4-D14	D14R, D14E, D14M, D14X, D14T, S08
57–59	05 / Supporting investigation	S23, S24, S33, S16

Navigate from a presentation section to its evidence

The slide range and source keys form a concise traceability index.

Slides	Section / diagnostic	Primary source keys
60–62	06 / Planned	S08, S02, S10
63–73	06 / Deferred	S02, S28, S09
74–82	07 / Appendix	S08, S10, S02, S23, S25, D06T, D06E + notes

Review the few decisions that affect enterprise adoption

The repository evidence supports a clear product story with explicit current limits.

Confirm the release

Validate the deployed diagnostic register and acceptance of recent D08/D11 changes.

Confirm the operating policy

Agree RCA conclusion approval, remediation ownership and fresh-data retest expectations.

Confirm enterprise readiness

Validate identity/access, portfolio-specific defaults and intended monitoring scope.

The next decision is a scoped product review supported by the source-linked evidence in this deck.